

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T40 · Angles in Polygons and Parallel Lines

With Answers

Questions with full worked solutions

■ 34 pages

FAMILY MAP

Angles in Polygons Parallel Lines

F01	Use regular and general polygon angles	11 Q 46 marks
F02	Use quadrilateral properties	2 Q 9 marks
F03	Use triangle angle properties	1 Q 4 marks
F04	Use basic angle and line facts	1 Q 5 marks

Question 14

4MA1-2H Q3

ORIGINAL QUESTION

4 marks

- 3 $ABCD$ is a trapezium.

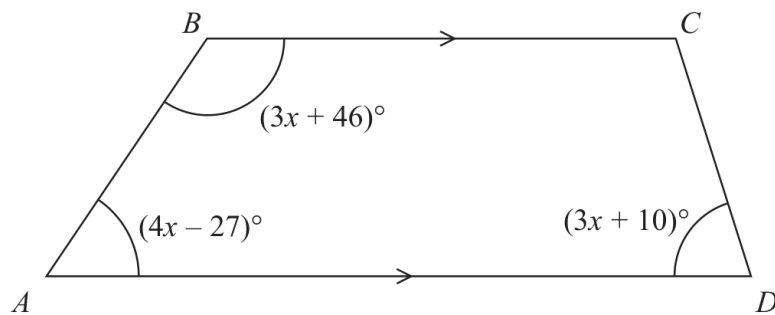


Diagram **NOT**
accurately drawn

BC is parallel to AD

Find the size of the largest angle inside the trapezium.

Worked Solution 14

Largest angle in a trapezium

MODULAR UNIT 1

4 marks

Use parallel-line angles

$$(4x - 27) + (3x + 46) = 180 \Rightarrow 7x + 19 = 180 \Rightarrow x = 23.$$

Evaluate the angles

$$3x + 46 = 115, \quad 3x + 10 = 79, \quad 4x - 27 = 65; \text{ the remaining angle is } 101.$$

Final answer

$$115^\circ$$

- 4 The diagram shows a triangle.

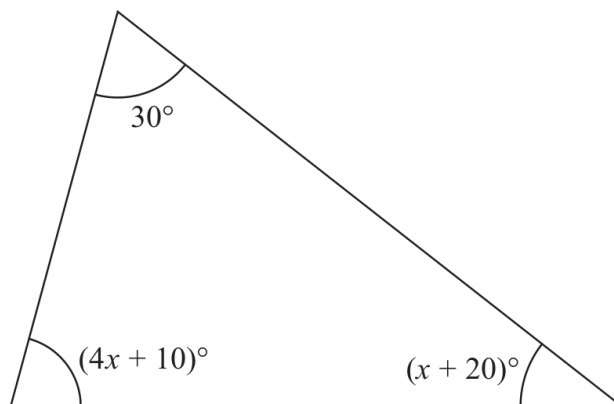


Diagram **NOT**
accurately drawn

Work out the value of x .

$x = \dots\dots\dots$

Worked solution

January 2020 | Question 4 | 4 marks

Family: Use triangle angle properties • Variant: Solve an algebraic triangle-angle chain with reasons

Method / working

1. Triangle angles sum to 180° .
2. $30 + (4x + 10) + (x + 20) = 180$.
3. $5x + 60 = 180$, so $5x = 120$.
4. $x = 24$.

Final answer: $x = 24$

5

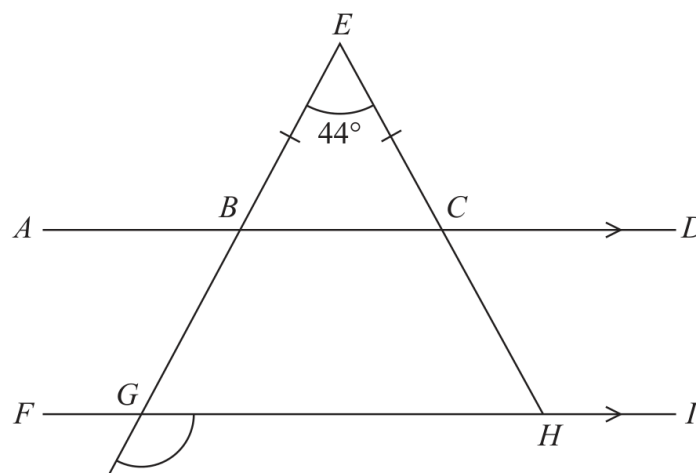


Diagram **NOT**
accurately drawn

$ABCD$ and $FGHI$ are parallel straight lines.
 $EBGJ$ and ECH are straight lines.

$BE = CE$
Angle $BEC = 44^\circ$

Work out the size of angle JGH .
Give a reason for each stage of your working.

o

Worked solution

June 2021 | Question 5 | 5 marks

Family: Use basic angle and line facts • Variant: Complete a multi-step line-angle reason chain

Method / working

1. $BE=CE$, so base angles of triangle BEC are $(180-44)/2=68^\circ$.
2. Use the straight line at B: the adjacent angle is $180-68=112^\circ$.
3. Transfer the angle through the parallel lines using corresponding/alternate angles.
4. Use the straight-line relation at G to reach $\text{angle JGH}=112^\circ$.
5. State the isosceles, straight-line, parallel and angle-sum reasons.

Final answer: 112°

3

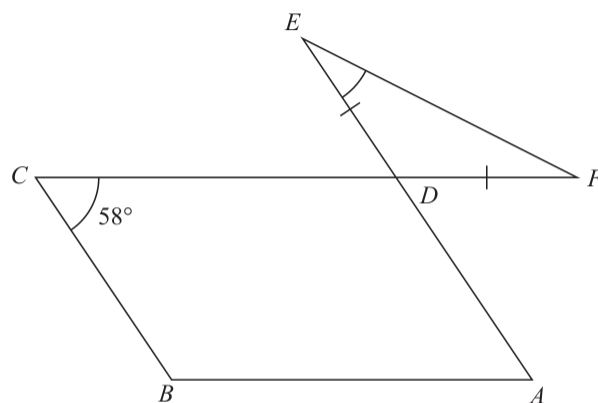


Diagram **NOT**
accurately drawn

The diagram shows a parallelogram $ABCD$ and an isosceles triangle DEF in which $DE = DF$

CDF and ADE are straight lines.

Angle $BCD = 58^\circ$

Work out the size of angle DEF .

Give a reason for each stage of your working.

(Total for Question 3 is 5 marks)

Answer reference | May-Nov 2020 | Question 3

Family: Use quadrilateral properties • Variant: Give standard reasons in a quadrilateral angle chain

3	$ADC = 180 - 58 (= 122)$ or $EDF = 122$ or $CDE = 58$ or $ADF = 58$			M1 may be seen marked on the diagram
	e.g. $DEF = 58 \div 2$ or $DEF = (180 - 122) \div 2$			M1 complete method to find angle DEF
		29		A1
			5	B2 dep on M2 for fully correct reasons for their method (B1 dep on M1 for one correct reason stated and used) e.g. <u>Allied angles</u> , <u>co-interior angles</u> , <u>Alternate angles</u> , <u>Corresponding angles</u> , <u>Vertically opposite angles</u> are equal (or Vertically <u>opposite angles</u> are equal), <u>Angles</u> on a straight <u>line</u> add up to 180° (or angles on a straight <u>line</u> add to 180°), Sum of <u>two angles</u> in a triangle are equal to <u>opposite exterior angle</u> , <u>Angles</u> in a <u>triangle</u> add up to 180° (or Angles in a <u>triangle</u> add up to 180°), Base angles in an <u>isosceles triangle</u> <u>Angles</u> in a <u>quadrilateral</u> add up to 360. (accept "4- sided shape" or parallelogram) <u>Opposite angles</u> of a <u>parallelogram</u> are equal
Total 5 marks				

Worked solution

May-Nov 2020 | Question 3 | 5 marks

Family: Use quadrilateral properties • Variant: Give standard reasons in a quadrilateral angle chain

Official final mark-scheme pages are embedded immediately after this question.

3

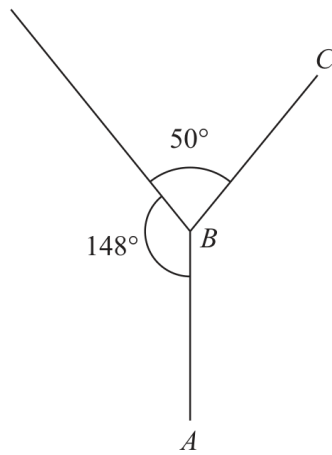


Diagram **NOT**
accurately drawn

AB and BC are two sides of a regular polygon with n sides.

Work out the value of n

Show your working clearly.

$n = \dots\dots\dots$

(Total for Question 3 is 4 marks)

Worked Solution 03

Interior and exterior angles of a regular polygon

UNIT 2
4 marks**Interior angle**

$$\angle ABC = 360^\circ - (148^\circ + 50^\circ) = 162^\circ.$$

Exterior angle and number of sides

$$e = 180^\circ - 162^\circ = 18^\circ, \quad n = \frac{360^\circ}{18^\circ} = 20.$$

Final answer

$$n = 20$$

Question 04

4MA1-1H Q5

ORIGINAL QUESTION

4 marks

- 5 In the diagram, $ABCDE$ is a regular pentagon.

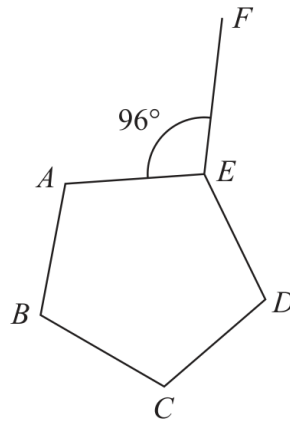


Diagram **NOT**
accurately drawn

Angle $AEF = 96^\circ$

Work out the size of the obtuse angle FED
Show your working clearly.

Worked Solution 04

Regular polygon angle

MODULAR UNIT 2

4 marks

Interior angle

$$(5 - 2)180^\circ / 5 = 108^\circ.$$

Around the point

$$\angle FED = 360^\circ - 96^\circ - 108^\circ = 156^\circ.$$

Final answer

$$156^\circ$$

Question 24

4MA1-2H Q11

ORIGINAL QUESTION

3 marks

- 11 Here is a 9-sided regular polygon $ABCDEFGHIJ$, with centre O

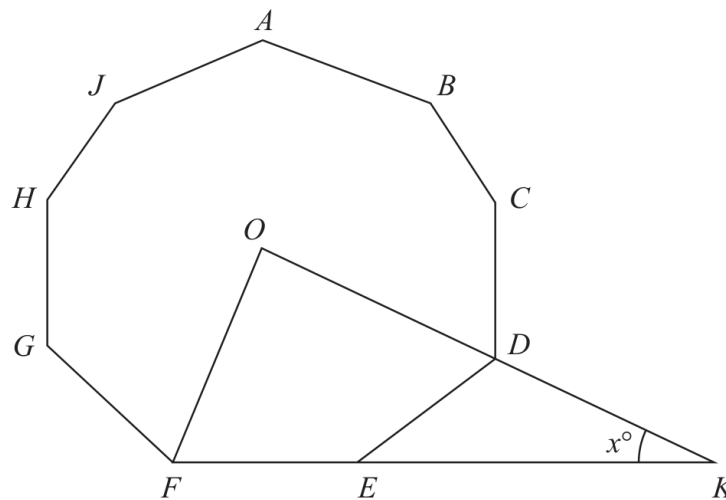


Diagram **NOT**
accurately drawn

ODK and FEK are straight lines.

Work out the value of x

$x = \dots\dots\dots$

Worked Solution 24

Angles in a regular nonagon

MODULAR UNIT 2

3 marks

Use the regular polygon angle

$$\text{Exterior angle} = 360/9 = 40^\circ, \quad \text{interior angle} = 140^\circ.$$

Use the straight lines

$$\angle OFK = \angle FOK = 80^\circ \text{ (isosceles triangle), and } \angle EDK = 110^\circ.$$

Read the small triangle

$$x = 180 - 110 - 40 = 30^\circ.$$

Final answer

$$30^\circ$$

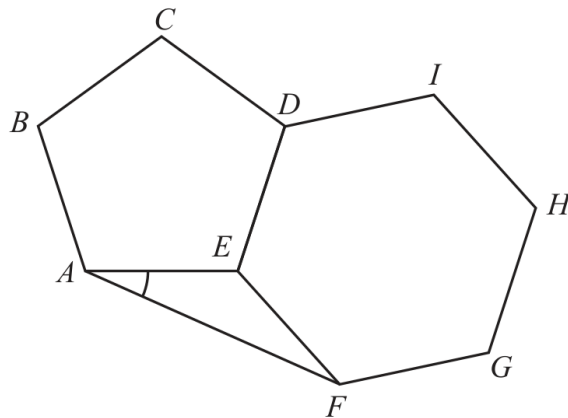
Original question 25

4MA1-2H Q11 M01

MODULAR UNIT 2

5 marks

11

Diagram **NOT**
accurately drawn*ABCDE* is a regular pentagon.*DEFGHI* is a regular hexagon.*AF* is a straight line.Work out the size of angle *EAF*

o

Worked solution 25

Angles in a regular pentagon and hexagon

MODULAR UNIT 2

5 marks

M01 – Angles in a regular pentagon and hexagon**Find the pentagon interior angle**

$$(5 - 2)180/5 = 108^\circ.$$

Find the hexagon interior angle

$$(6 - 2)180/6 = 120^\circ.$$

Find the obtuse angle

$$\angle AEF = 360 - 108 - 120 = 132^\circ.$$

Use the isosceles triangle

$$\angle EAF = (180 - 132)/2 = 24^\circ.$$

Final answer

24 degrees

Original question 05

4MA1-1H Q4 Q4

ORIGINAL QUESTION**4 marks**

- 4 The diagram shows parts of three regular polygons, **A**, **B** and **C**, meeting at a point.

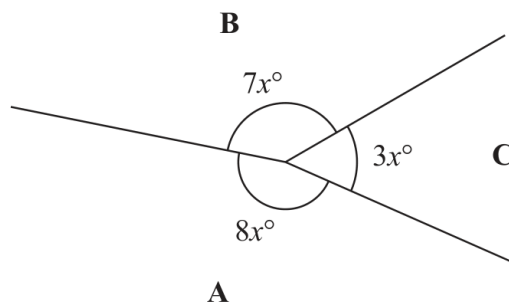


Diagram **NOT**
accurately drawn

Polygon **B** has n sides.

Work out the value of n .

$n =$

Worked solution 05

Chapter: Angles in Polygons & Parallel Lines

MODULAR UNIT 2

4 marks

Q4 – Chapter: Angles in Polygons & Parallel Lines

Family: Use regular and general polygon angles • Variant: Combine several regular polygons around lines or a point

Method

The three angles around the point give $7x + 3x + 8x = 360$ degrees.

Why: Angles around one point total 360 degrees.

Method

$18x = 360$, so $x = 20$ degrees.

Why: Solve the linear equation.

Method

Polygon B has interior angle $7x = 140$ degrees, so its exterior angle is $180 - 140 = 40$ degrees.

Why: An interior angle and its corresponding exterior angle are supplementary.

Method

For a regular polygon, $n = 360/40 = 9$.

Why: The exterior angles of any polygon total 360 degrees.

Final answer

number of sides: $n = 9$

Original question 19

4MA1-1H Q23 Q23

ORIGINAL QUESTION

6 marks

23 A polygon has n sides, where $n > 5$

When arranged in order of size, starting with the largest number, the sizes of the interior angles of the polygon, in degrees, are the terms of an arithmetic sequence.

Here are the first five terms of this sequence.

177 175 173 171 169

Find the value of n

Show clear algebraic working.

Question 23 continued

$n =$

Worked solution

Family: Use regular and general polygon angles • Variant: Solve an irregular polygon angle or side-count

METHOD / WORKING

MODULAR UNIT 2 • JUNE 2022 DETAILED SOLUTIONS
Worked solution 19 MODULAR UNIT 2
Chapter: Angles in Polygons & Parallel Lines 6 marks
Q23 – Chapter: Angles in Polygons & Parallel Lines
Family: Use regular and general polygon angles • Variant: Solve an irregular polygon angle or side-count
problem
Method
The interior-angle sequence has first term 177 and common difference $d = -2$.
Why: Successive listed angles decrease by 2 degrees.
Method
$Itsumis S_n = n/2[2(177) + (n - 1)(-2)].$
Why: Apply the arithmetic-series sum formula.
Method
The interior-angle sum of an n -sided polygon is $180(n - 2)$, so $n/2[354 - 2(n - 1)] = 180(n - 2)$.
Why: Equate the sequence sum to the polygon angle sum.
Method
$Simplifying gives n^2 + 2n - 360 = 0.$
Why: Expand and collect all terms on one side.
Method
$n^2 + 2n - 360 = (n - 18)(n + 20), so n = 18 or n = -20.$
Why: Factorise the quadratic and use the zero-product rule.
Method
The number of sides is positive and $n > 5$, hence $n = 18$.
Why: Apply the domain condition for a polygon.
Final answer
number of sides: $n = 18$

FINAL ANSWER

Original question 25

4MA1-2H Q11

ORIGINAL QUESTION**4 marks**

- 11 The diagram shows a regular 10-sided polygon, $ABCDEFGHIJ$

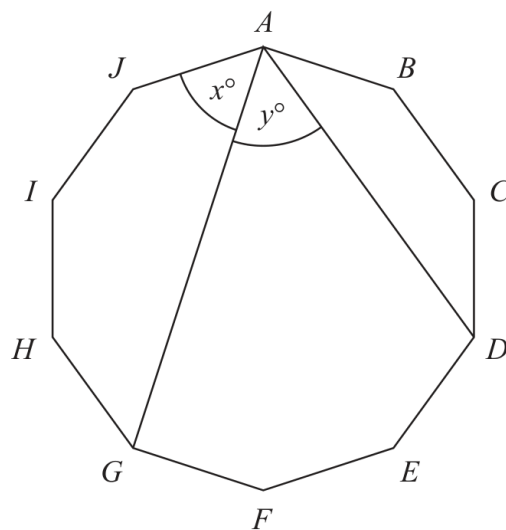


Diagram **NOT**
accurately drawn

Show that $x = y$

Worked solution 25

Chapter: Angles in Polygons & Parallel Lines

MODULAR UNIT 2

4 marks

Q11 – Chapter: Angles in Polygons & Parallel Lines

Family: Use regular and general polygon angles • Variant: Find interior or exterior angle of a regular polygon

Method

The exterior angle of a regular decagon is $360/10=36^\circ$, so each interior angle is 144° .

Why: Use the polygon angle formula.

Method

In triangle formed at A, $x=144-90=54^\circ$.

Why: Use the right/straight-angle relationship shown.

Method

The corresponding angle y is also 54° from the equal-angle/isosceles geometry.

Why: Use the symmetry of the regular decagon and the marked triangles.

Method

Thus $x=y=54^\circ$.

Why: State the required result.

Final answer

angles: $x=y=54^\circ$

- 5 The diagram shows a regular octagon $ABCDHIJK$ and a pentagon $DEFGH$.

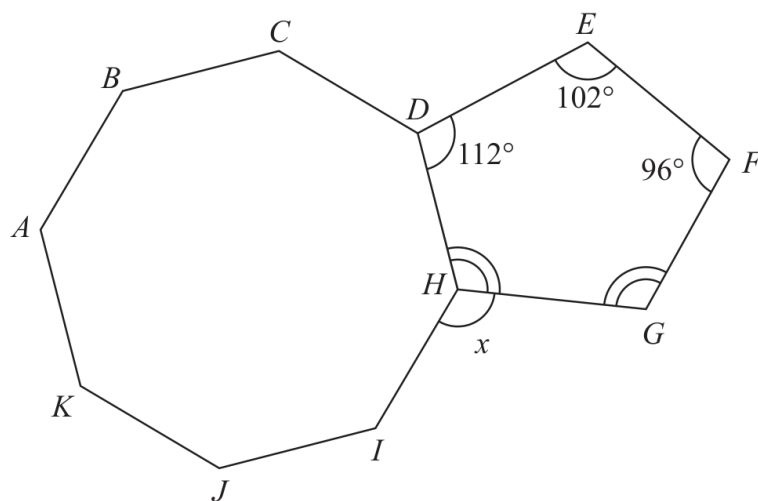


Diagram **NOT**
accurately drawn

Angle $GHD = \text{angle } FGH$.

Work out the size of the angle marked x .
Show your working clearly.

○

Worked solution

November 2021 | Question 5 | 5 marks

Family: Use regular and general polygon angles | Variant: Combine several regular polygons around lines or a point

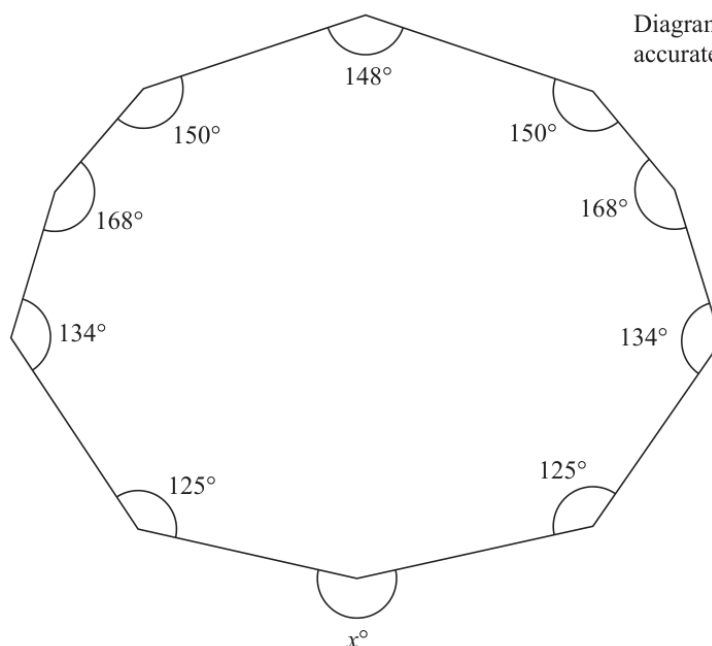
METHOD / WORKING

1. For pentagon $DEFGH$, the interior angle sum is
2. $(5 - 2) \times 180 = 540^\circ$
3. Let
4. $\angle GHD = \angle FGH = a$
5. Then
6. $112 + 102 + 96 + a + a = 540$
7. $2a = 230$
8. $a = 115^\circ$
9. A regular octagon has interior angle
10. $\frac{(8 - 2) \times 180}{8} = 135^\circ$
11. At H , angles around a point add to 360° :
12. $x = 360^\circ - 135^\circ - 115^\circ = 110^\circ$
13. Answer: 110° .

FINAL ANSWER

 110° .

8 Here is a 10-sided polygon.



Work out the value of x .

$x = \dots\dots\dots$

(Total for Question 8 is 4 marks)

Answer reference | May-Nov 2020 | Question 8

Family: Use regular and general polygon angles • Variant: Find the interior-angle sum of a polygon

8	$(10 - 2) \times 180$ oe (= 1440) or $(6 - 2) \times 180$ oe (= 720)		4	M1	for a method to find the sum of the interior angles of a decagon or a hexagon
	'1440' - 148 - 2×150 - 2×168 - 2×134 - 2×125 (=138) or '1440' - 1302 (= 138) or '720' - $148 \div 2$ - 150 - 168 - 134 - 125 (= 69) or '720' - 651 (= 69)			M1	Allow omission of one angle
	$360 - '138'$ or $360 - 2 \times '69'$			M1	
		222		A1	
	Alternative method (exterior angles)				
	$360 - 2 \times (180 - 125) - 2 \times (180 - 134) - 2 \times (180 - 168) - 2 \times (180 - 150) - (180 - 148)$ or $360 - 2 \times 55 - 2 \times 46 - 2 \times 12 - 2 \times 30 - 32$		4	M2	If not M2 then award M1 for at least 3 or (180 - 125) , (180 - 134) , (180 - 168) , (180 - 150), (180 - 148) or at least 3 of 55, 46, 12, 30, 32
	$180 + '42'$			M1	
		222		A1	
					Total 4 marks

Worked solution

May-Nov 2020 | Question 8 | 4 marks

Family: Use regular and general polygon angles • Variant: Find the interior-angle sum of a polygon

Solution

The sum of the interior angles of a 10-sided polygon is

$$(10-2) \times 180 = 1440^\circ$$

The sum of the 9 given interior angles is

$$150 + 168 + 134 + 125 + 125 + 134 + 168 + 150 + 148 = 1302^\circ$$

So the missing interior angle is

$$1440^\circ - 1302^\circ = 138^\circ$$

The angle x is the reflex angle outside the polygon at this vertex:

$$x = 360^\circ - 138^\circ = 222^\circ$$

Answer: 222° .

- 4 The diagram shows a pentagon.

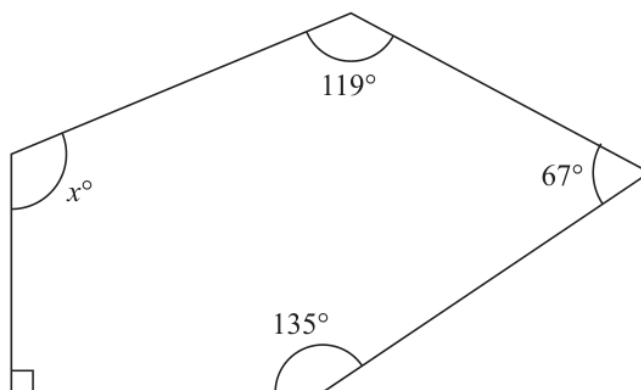


Diagram **NOT**
accurately drawn

Work out the value of x

$x =$

(Total for Question 4 is 3 marks)

Answer reference | January 2023 | Question 4

Family: Use regular and general polygon angles • Variant: Find the interior-angle sum of a polygon

4	$3 \times 180 (= 540)$ or $360 - [(180 - 90) + (180 - 135) + (180 - 67) + (180 - 119)] (= 51)$ or $360 - (90 + 45 + 113 + 61) (= 51)$		3	M1
	$90 + 135 + 67 + 119 + x = "540"$ oe $411 + x = "540"$ oe or $"540" - (90 + 135 + 67 + 119)$ or $3 \times 180 - (90 + 135 + 67 + 119)$ oe or $540 - 411$ or $180 - "51"$ oe			M1
	Correct answer scores full marks (unless from obvious incorrect working)	129		A1
				Total 3 marks

Worked solution

January 2023 | Question 4 | 3 marks

Family: Use regular and general polygon angles • Variant: Find the interior-angle sum of a polygon

Official final mark-scheme pages are embedded immediately after this question.

12

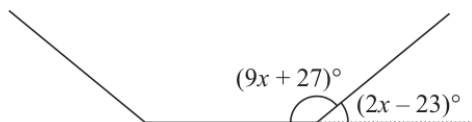


Diagram **NOT**
accurately drawn

The diagram shows part of a regular n -sided polygon with

an interior angle of $(9x + 27)^\circ$

an exterior angle of $(2x - 23)^\circ$

Work out the value of n

$n =$

(Total for Question 12 is 4 marks)

Worked solution

November 2025 | Question 12 | 4 marks

Family: Use regular and general polygon angles • Variant: Solve an irregular polygon angle or side-count problem

Solution

The interior angle and exterior angle of a regular polygon add to 180° :

$$(9x+27)+(2x-23)=180$$

$$11x+4=180$$

$$x=16$$

The exterior angle is

$$2x-23=2(16)-23=9^\circ$$

For a regular polygon,

$$n=(360)/(9)=40$$

Answer: $n=40$.